Description of Additional Supplementary Files
Self-oscillating synchronematic colloids
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File Name: Supplementary Video 1, 1_fluid.mp4
Description: Large Fluid Cluster. Particle dynamics within a large fluid cluster, in which the particle density is approximately uniform throughout the imaging region (Figs. 1d and 2a). Video is acquired at 3000 frames per second (fps) and played back at 30 fps. The imaging region is 1280x800 px^2 = 1702x1064 μm^2 in size. Particles are annotated by colored ellipsoids that denote the oscillation phase and orientation.

File Name: Supplementary Video 2, 2_crystals.mp4
Description: Synchronematic Crystals. Particle dynamics within three synchronematic crystals (Fig. 1e). Video is acquired at 3000 fps and played back at 30 fps for a duration of 400 frames (133 ms real time). The imaging region is 420x264 px^2 = 559x351 μm^2 in size. The upper frame shows the raw images; the lower frame is annotated by colored ellipsoids that denote the oscillation phase and orientation.

File Name: Supplementary Video 3, 3_fluid_gradient.mp4
Description: Small Fluid Cluster. Particle dynamics within a smaller fluid cluster, in which the particle density is varies across the imaging region (Figs. 2e,f). Video is acquired at 3000 fps and played back at 30 fps. The imaging region is 1280x800 px^2 = 1702x1064 μm^2 in size. Particles are annotated by colored ellipsoids that denote the oscillation phase and orientation.

File Name: Supplementary Video 4, 4_single_crystal.mp4
Description: Single Synchronematic Crystal. Particle dynamics within a single synchronematic crystal containing $N \approx 110$ particles (Figs. 3a-c). Video is acquired at 3000 fps and played back at 15 fps for a duration of 400 frames (133 ms real time). The imaging region is 200x200 px^2 = 266x266 μm^2 in size. The images are annotated to show: (top, left) colored disks representing the hexatic order parameter ψ_6; (top, right) colored triangles indicating the instantaneous particle velocity $\mathbf{v}$; (lower, right) colored ellipsoids denoting the oscillation phase θ and orientation angle χ; (lower, left) colored disks representing the phase-angle order parameter $S_{\theta\chi}$ based on six nearest neighbors.

File Name: Supplementary Video 5, 5_single_crystal_sim.mp4
Description: Single Synchronematic Crystal (SD simulation). Simulated particle dynamics within a single synchronematic crystal containing $N = 70$ particles (Figs. 3d-f). The total simulation time is 44 ms. The images are annotated to show: (top, left) colored disks representing the hexatic order parameter ψ_6; (top, right) colored triangles indicating the instantaneous particle velocity $\mathbf{v}$ (lower, right) colored ellipsoids denoting the oscillation phase θ and orientation angle χ; (lower, left) colored disks representing the phase-angle order parameter $S_{\theta\chi}$ based on six nearest neighbors.

File Name: Supplementary Video 6, 6_linear_to_circular.mp4

Description: Linear to Circular Order. Transition from linear to circular synchronematic ordering extracted from experiments (Fig. 4a). Video is acquired at 3000 fps and played back at 15 fps for a duration of 720 frames (240 ms real time). The imaging region is 260x260 px^2 = 346x346 μm^2 in size. The images are annotated to show: (top, left) colored disks representing the hexatic order parameter ψ_6; (top, right) colored triangles indicating the instantaneous particle velocity **v**; (lower, right) colored ellipsoids denoting the oscillation phase θ and orientation angle χ; (lower, left) colored disks representing the phase-angle order parameter $S_{\theta\chi}$ based on six nearest neighbors.

File Name: Supplementary Video 7, 7_elongated.mp4
Description: Linear to Circular Order (SD simulation). Transition from linear to circular synchronematic ordering extracted from simulations (Fig. 4b). The cluster contains N = 70 oscillators, and the video duration is 440 ms of simulated time. The images are annotated to show: (top, left) colored disks representing the hexatic order parameter ψ_6; (top, right) colored triangles indicating the instantaneous particle velocity **v**; (lower, right) colored arrows denoting the oscillation phase θ and orientation angle χ; (lower, left) colored disks representing the linear nematic order parameter S_l based on six nearest neighbors.

File Name: Supplementary Video 8, 8_disassembly.mp4
Description: Crystal Disassembly. Particle ejection and catastrophic disassembly of two synchronematic crystals (played sequentially). Video is acquired at 3000 fps and played back at 30 fps for a duration of 600 frames (200 ms real time). The imaging region is 200x200 px^2 = 266x266 μm^2 in size. Particles are annotated by colored triangles indicating the instantaneous particle velocity **v**.

File Name: Supplementary Video 9, 9_circular_reduced_order_N210.mov
Description: Small Synchronematic Crystal (swarmalator simulation). Reduced-order simulation dynamics for a crystalline cluster exhibiting global synchronematic order with circular alignment. The number of particles is N = 210; the rotational diffusion coefficient D_χ = 0.001. The integration time step is Δt = 0.1; snapshots for the video are taken every 10 time steps.

File Name: Supplementary Video 10, 10_circular_reduced_order_N500.mov
Description: Large Synchronematic Crystal (swarmalator simulation). Reduced-order simulation dynamics for a crystalline cluster exhibiting local synchronematic order. The number of particles is N = 500; the rotational diffusion coefficient D_χ = 0.001. The the integration time step is Δt = 0.1; snapshots for the video are taken every 10 time steps.

File Name: Supplementary Video 11. 11_random_reduced_order.mov
Description: Bulk Fluid (swarmalator simulation). Reduced-order simulation dynamics for a system with N = 1600 particles, area fraction ρ = 0.3, and rotational diffusion coefficient $D\chi$ = 0.001. Even with such small noise levels, the system exhibits finite

correlation lengths. The the integration time step is $\Delta t = 0.1$; snapshots for the video are taken every 10 time steps.

File Name: Supplementary Video 12. 12_lattice_reduced_order.mov
Description: Bulk Crystal (swarmalator simulation). Reduced-order simulation dynamics for a system with $N = 8100$ particles, area fraction $\rho = 0.3$, and rotational diffusion coefficient $D_X = 0.001$. The system exhibits defects in phase and orientation. The the integration time step is $\Delta t = 0.1$; snapshots for the video are taken every 10 time steps.